

HMWK 2

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Introduction

The purpose of this assignment was to implement and compare two methods for classifying the three Iris flower species: a two stage binary logistic regression classifier and a multinomial softmax regression classifier. Both approaches were trained using gradient descent and evaluated using accuracy and confusion matrices.

Part 1 – Binary Logistic Regression

We trained the binary logistic regression model to distinguish setosa apart from the other two Iris species using a learning rate of 0.5, regularization parameter of 0.1 and 2,000 iterations. The model achieved 100% training accuracy and 100% testing accuracy. The model was able to correctly separate setosa flowers from the other two species.

Figure 1: Binary Logistic Regression Training Loss

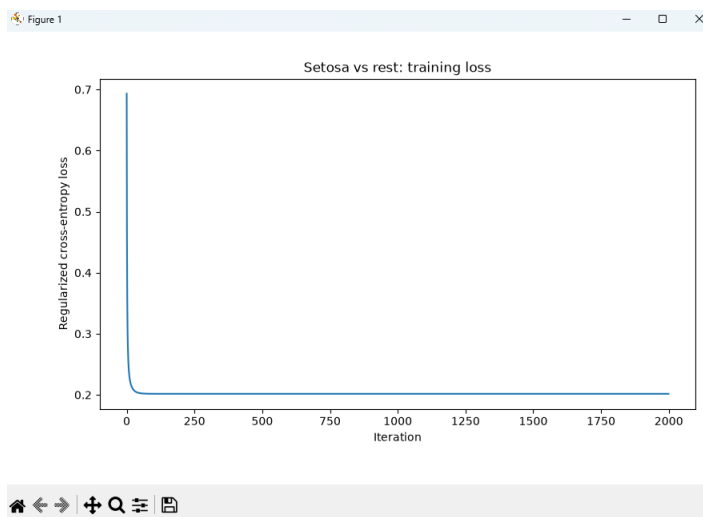


Figure 1. Regularized cross-entropy loss over 2,000 iterations

The training loss decreased from roughly 0.69 to 0.20, with most of the improvement occurring during the first 100 iterations. Afterward, the loss stabilized around 0.20. These results show that gradient descent successfully minimized the loss and that the model converged.

Part 2 – Two Binary Classifiers vs. Softmax Regression

2.1 Two Stage Classifier

The two stage classifier uses two binary logistic regression models. The first determines where a flower is setosa or not, while the second distinguishes between versicolor and virginica. The second is only trained on non-setosa samples. The probabilities from both classifiers are combined using the chain rule to produce probabilities for all three species. These probabilities add up to 1 for each sample.

2.2 Softmax Regression

Softmax regression uses a single model to classify all three Iris species simultaneously. Using the softmax function, it calculates probabilities for each species and selects the class with the highest probability. The model uses cross-entropy loss, L2 regularization, and batch gradient descent. The maximum value of each row is subtracted before exponentiation. The model was trained using a learning rate of 0.5, regularization parameter of 0.01 and 3,000 iterations.

2.3 Model Comparison

Both models were trained and evaluated using the same dataset split.

Figure 2: Results

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Two-stage: train acc = 0.9810, test acc = 0.9111, test log loss = 0.1742
macro precision = 0.9111, macro recall = 0.9111
Test confusion matrix (rows = true class, cols = predicted class; order: ['setosa', 'versicolor', 'virginica']):
[[15  0  0]
 [ 0 13  2]
 [ 0  2 13]]

Softmax: train acc = 0.9810, test acc = 0.9333, test log loss = 0.1923
macro precision = 0.9345, macro recall = 0.9333
Test confusion matrix (rows = true class, cols = predicted class; order: ['setosa', 'versicolor', 'virginica']):
[[15  0  0]
 [ 0 13  2]
 [ 0  1 14]]
```

Figure 2. Displayed results from running the file

Both models achieved the same training accuracy of 98.10%. However, softmax regression performed slightly better on the test set, achieving 93.33% accuracy compared to 91.11% for the two-stage classifier. Softmax made three incorrect while the two-stage classifier made four. Surprisingly, we observed the two-stage classifier achieved a lower test log loss, indicating that its predicted probabilities performed better despite having slightly lower classification accuracy.

Part 3 – Short Answer Interpretation

3.1

The classes that were confused were versicolor and virginica. The two-stage classifier incorrectly classified versicolor samples as virginica and two virginica samples as versicolor,

whereas softmax made the same two mistakes with versicolor but only incorrectly classified one virginica sample. Neither model incorrectly classified any setosa samples.

Figure 3: Iris Training Set by Species

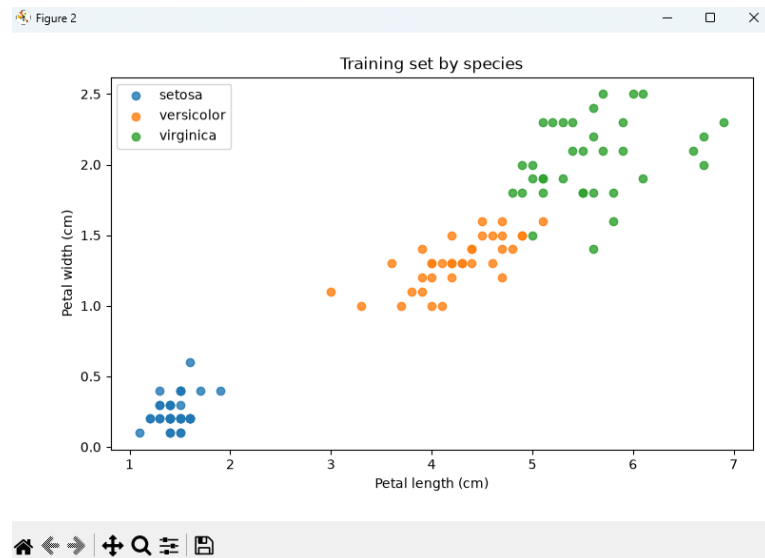


Figure 3. Petal length versus petal width for the training samples, covered by Iris species

Setosa can be seen as its own separate cluster, compared to versicolor and virginica which are much closer together and overlap slightly around petal lengths of 4.8-5.0cm and pedal widths of 1.4-1.6cm. This overlap helps explain why both models occasionally confuse these two species.

3.2 Chain Rule Explanation

The two-stage classifier cannot compare the outputs of its two binary classifiers because they represent different probabilities. The first classifier predicts the probability that a flower is setosa, while the second predicts the probability that a flower is virginica given that it is not setosa.

The factor $(1 - q_1)$ represents the probability that a flower is not a setosa. Multiplying this factor by the second classifier's probabilities accounts for that condition and produces probabilities that can be compared across all three species.

3.3 Model Recommendation

Based on the results, we recommend softmax regression for this classification problem. Although both models achieved the same training accuracy at 98.10%, softmax achieved a higher testing accuracy at 93.33% compared to 91.11% for the two-stage classifier. Softmax also made one less classification error correctly identifying 14 virginica samples against 13 for the two-stage model. The two stage model did achieve a slightly lower test log loss, but softmax

performed better in terms of overall classification accuracy. Softmax is also able to classify all three species with a single model instead of combining two separate classifiers. Overall, softmax regression appears to be the better choice for this dataset.